

3.2.3 Machine Learning Dataset Collection and Processing

- Collect and preprocess datasets containing features such as inventory size, fragile item count, distance, and demand patterns.
- Implement **data-cleaning and normalization techniques** to improve ML accuracy.
- Split the dataset into **training, testing, and validation sets** to avoid overfitting.

3.2.4 Implementation of ML-Pipeline

- Integrate a **classification model** for package recommendation (e.g., predicting the most suitable mover package).
- Deploy a **regression model** for dynamic pricing, accounting for variables like traffic, distance, time of booking, and demand fluctuations.
- Ensure models are **integrated into the backend** for real-time predictions.

3.2.5 AI Chatbot Integration

- Develop a **chatbot using NLP techniques** to handle common user queries such as booking confirmation, tracking status, and pricing.
- Predefine **conversational flows** for smooth interactions while also allowing escalation to human support for complex issues.
- Train the chatbot to improve over time with **user feedback and conversational logs**.

3.2.6 Value-Added Features

- **Real-Time Tracking:** Use GPS-based updates to provide users with live information about the location of their belongings.
- **Insurance Options:** Offer users the ability to insure fragile or valuable goods to reduce financial risk in case of damage.
- **Review and Rating System:** Allow customers to provide feedback, thereby improving transparency and helping refine service quality.

Together, these objectives transform SHIFTY into not just another moving app but a **data-driven, customer-first solution** that enhances the overall moving experience.